

Half Yearly Examination 2024

Class XI

Mathematics (Code- 041)

Time: 3 Hours

MM-80

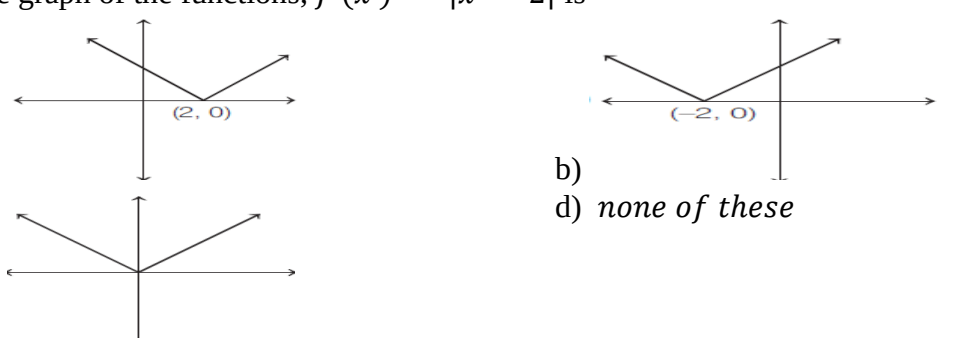
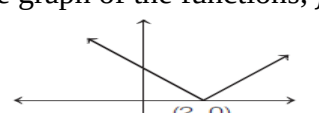
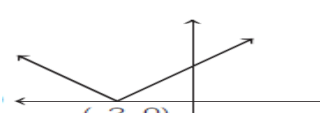
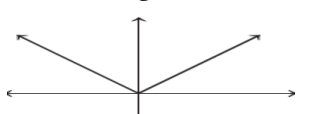
Instructions: -

1. This Question paper contains - five sections A, B, C, D and E. Each section is compulsory. However, there are internal choices in some questions.
2. Section A has 18 MCQ's and 02 Assertion-Reason based questions of 1 mark each.
3. Section B has 5 Very Short Answer (VSA)-type questions of 2 marks each.
4. Section C has 6 Short Answer (SA)-type questions of 3 marks each.
5. Section D has 4 Long Answer (LA)-type questions of 5 marks each.
6. Section E has 3 source based/case based/passage based/integrated units of assessment (4 marks each) with sub parts.

SECTION A

(Multiple Choice Questions)

Each question carries 1 mark

1.	Which of the following is a singleton set? a) $\{x: x < 4, x \in N\}$ b) $\{x: x < -4, x \in N\}$ c) $\{x: x^2 = 4, x \in N\}$ d) $\{x: x^2 = 4, x \in R\}$	1
2.	The number of subsets of a set containing 4 elements is a) 4 b) $2^4 - 1$ c) 4^2 d) 2^4	1
3.	The r^{th} term of a GP 3, 9, 27, ... is a) 3^n b) 3^{n-1} c) 3^{n+1} d) 3^{n-2}	1
4.	8^{th} term from the end of the sequence 3, 6, 12, 25^{th} term is a) 393216 b) 393206 c) 313216 d) 303216	1
5.	The value of $\tan(-1575^\circ)$ is: a) $\frac{1}{2}$ b) 1 c) -1 d) 0	1
6.	The Value of $2\sin^2 \frac{\pi}{6} + \operatorname{cosec}^2 \frac{7\pi}{6} \cos^2 \frac{\pi}{3}$ is: a) $\frac{1}{2}$ b) $-\frac{1}{2}$ c) $\frac{3}{2}$ d) 1	1
7.	$40^\circ 20'$ into radian is a) $\frac{121\pi}{540}$ b) $\frac{120\pi}{540}$ c) $\frac{221\pi}{540}$ d) $\frac{41\pi}{540}$	1
8.	If $^{15}\text{Cr} : ^{15}\text{Cr}_{-1} = 11:5$ then r equals a) 11 b) 15 c) 5 d) 4	1
9.	A polygon has 35 diagonals. Find the number of its sides. a) 10 b) 7 c) 16 d) 70	1
10.	Let $f = \{(1, 1), (2, 3), (0, -1), (-1, -3)\}$ be a function from Z to Z defined by $f(x) = mx + c$. the value of c is a) -11 b) 11 c) 1 d) -1	1
11.	The graph of the functions, $f(x) = x - 2 $ is  a)  b)  c)  d) none of these	1
12.	If $(2 + 5i)z = (3 - 7i)$, then z is equal to	

	a) $1 + i$ b) $1 - i$ c) $-1 + i$ d) $-1 - i$	1
13.	The value of $\left[i^{19} + \left(\frac{1}{i}\right)^{25}\right]^2$ is equal to a) 4 b) -4 c) i d) $-i$	1
14.	If $ x - 1 > 5$, then a) $x \in (-4, 6)$ b) $x \in [-4, 6]$ c) $x \in (-\infty, -4) \cup (6, \infty)$ d) $x \in [-\infty, -4) \cup [6, \infty)$	1
15.	If $-3x + 17 < -13$, then a) $x \in (10, \infty)$ b) $x \in [10, \infty)$ c) $x \in (-\infty, 10]$ d) $x \in [-10, 10)$	1
16.	The coefficient of the 3rd term in the expansion of $(2 + 3x)^4$ is: a) 5! b) 6 c) 216 d) 8!	1
17.	If the coefficient of x^2 and x^3 in the expansion of $(3 + mx)^9$ are equal, then the value of m is a) $-9/7$ b) $-7/9$ c) $9/7$ d) $7/9$	1
18.	The coefficient of x^3y^4 in $(2x + 3y^2)^5$ is a) 360 b) 720 c) 240 d) 1080	1

ASSERTION-REASON BASED QUESTIONS

In the following questions, a statement of assertion (A) is followed by a statement of Reason (R). Choose the correct answer out of the following choices.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true but R is not the correct explanation of A.
 (c) A is true but R is false.
 (d) A is false but R is true.

19.	Assertion (A): The power set of the set $\{1, 2\}$ is the set $\{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. Reason (R): The power set is set of all subsets of the set.	1
20.	Assertion (A): Sum of the series $-\frac{3}{4}, \frac{3}{16}, -\frac{3}{64}, \dots$ is $-\frac{2}{3}$. Reason (R): If a_1, a_2, a_3 are in GP, then $\frac{a_2}{a_1} = \frac{a_3}{a_2}$.	1

SECTION B

This section comprises of very short answer type-questions (VSA) of 2 marks each

21.	Two finite sets have m and n elements. The total number of subsets of first set is 56 more than the total number of subsets of the second set. Find the value of m and n.	2
22.	Find the sum of 0.15, 0.015, 0.0015, ... up to 20 terms OR If A.M. and G.M. of two positive numbers a and b are 10 and 8, respectively, find the numbers.	2
23.	A horse is tied to a post by a rope. If the horse moves along a circular path always keeping the rope tight and describes 88 m when it has traced out 72° at the centre, find the length of the rope.	2
24.	Solve the inequality and show the graph of the solution on number line. $\frac{2x-3}{4} + 9 \geq 3 + \frac{4x}{3}$ OR Solve the inequality: $\frac{x}{2x+1} \geq \frac{1}{4}$.	2
25.	Ravi scored 70 and 75 marks in the first two-unit test. Calculate the minimum marks he should get in the third test to have an average of at least 60 marks.	2

SECTION C

(This section comprises of short answer type questions (SA) of 3 marks each)

26.	Draw the graph of $f(x) = \operatorname{cosec} x$, also write Domain and range of the $f(x)$.	3
27.	The English alphabet has 5 vowels and 21 consonants. How many words with two different vowels and 2 different consonants can be formed from the alphabet? OR Find the number of words with or without meaning which can be made using all the letters of the word AGAIN. If these words are written as in a dictionary, what will be the 50th word?	3
28.	Given 5 different green dyes, four different blue dyes and three different red dyes, find the number of combinations of dyes which can be chosen taking at least one green and one blue dye.	3
29.	Draw graph of greatest integer function. Also write Domain and Range of the greatest integer function. OR Find the domain and range of the following real functions: $f(x) = \sqrt{4 - x^2}$	3
30.	Find the modulus and conjugate of $\frac{(2+i)^2}{3+i}$ OR If z_1, z_2 are $1 - i$ and $-2 + 4i$ respectively find $\operatorname{Im}\left(\frac{z_1 z_2}{z_1}\right)$	3
31.	The sum of two numbers is 6 times their geometric mean, show that numbers are in the ratio $(3 + 2\sqrt{2}) : (3 - 2\sqrt{2})$	3

SECTION D

(This section comprises of long answer-type questions (LA) of 5 marks each)

32.	Prove that: $\sin x + \sin 2x + \sin 4x + \sin 5x = 4 \sin 3x \cos \frac{x}{2} \cos \frac{3x}{2}$ OR Prove that: $\frac{(\cos x - \cos 3x)(\sin 8x + \sin 2x)}{(\sin 5x - \sin x)(\cos 4x - \cos 6x)} = 1$	5
33.	If $\tan x = \frac{5}{12}$, x lies in quadrant II. Find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$.	5
34.	If $(x + iy)^3 = u + iv$, then show that $\frac{u}{x} + \frac{v}{y} = 4(x^2 - y^2)$.	5
35.	Find the value of $(a^2 + \sqrt{a^2 - 1})^4 + (a^2 - \sqrt{a^2 - 1})^4$ OR Show that $2^{4n+4} - 15n - 16$, where $n \in \mathbb{N}$ is divisible by 225.	5

SECTION E

(This section comprises of 3 case-study/passage-based questions of 4 marks each with two sub-parts. First two case study questions have three sub parts (i), (ii), (iii) of marks 1, 1, 2 respectively. The third case study question has four sub parts 01 marks each.)

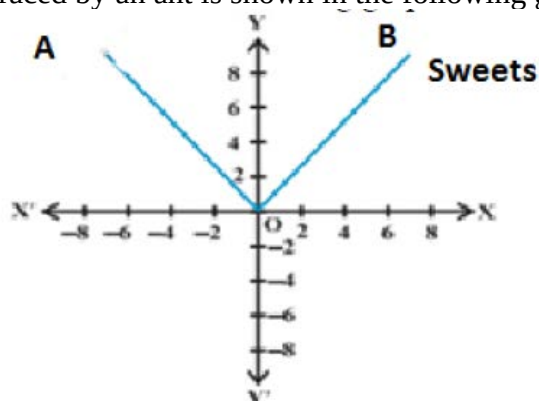
36.	Read the Case study given below and answer the questions. Independence Day is celebrated annually on 15 august as a public holiday in India. Answer the following questions which are based on the word 'Independence'	
-----	--	--



- Find the number of arrangements of the letters of the word Independence.
- In how many of these arrangements do all vowels come together.
- Find the number of arrangements of the letters when words begin with I and end in P?

1
1
2

37. A is the anthills of an ant, at B some sweets are there and ant wants to reach at B. The path traced by an ant is shown in the following graph:

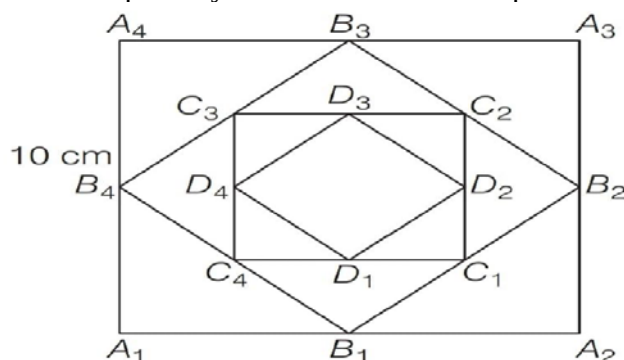


On the basis of the above graph find the following:

- When ordinate is 6 then find abscissa
- Which axis is line of symmetry for the graph?
- Write the function for the graph along with domain and range.

1
1
2

38. A student of class XI draws a square of side 10 cm. Another student joins the mid-points of this square to form a new square. Mid points of the sides of new square are joined to form another square by another student. This process is continued indefinitely.



On the basis of the above information, answer following questions:

- What is the perimeter of fourth square?
- Find the length of side of 10th square.

2
2